Lesson plan · for the teacher

How AI models work: unit check

AI · Level 3 · Unit check

Time	Materials
40 min	Board and chalk; The quiz sheet (one per team)

Learning goals

- One neuron
- How a network learns
- Train, test, trust?
- Inside a language model

Activity: Team quiz

1. Warm-up (5 min): Ask for one word that sums up each of the four lessons, and write them on the board.
2. Team quiz (15 min): In teams of four, read each question aloud; teams agree on one answer and write it on their sheet. Reveal the answer after each question, and ask one team to explain why.
3. Explain (8 min): Find the two questions most teams got wrong. Each team explains one of them on the board, using a drawing (a neuron, a hill, a test set or the loop).
4. Claim check (7 min): Read out one real AI claim (from an advert, an app or the news). Teams ask the three questions: tested on whom, on which data, with which mistakes? Agree what you would still need to know.
5. Wrap-up (5 min): Exit ticket: each person writes one idea from this unit they will use this week.

Print the quiz sheet (one per team) and read the other questions aloud from this check. Find one real AI claim to check.

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- Which AI claim in your school deserves the three questions?

Exit ticket

- One idea I will use, and when I will use it

Try it in class tomorrow

Choose one AI tool your school uses or is being offered, and ask its supplier the three questions: tested on whom, on which data, with which mistakes? Share the answers with a colleague.

Worksheet · for pupils

How AI models work: unit check

AI · Level 3 · Unit check

Name _____ Class _____ Date _____

1 **A neuron has inputs 1 and 1, weights 2 and −3, and a bias of 2. What does it say?**

◯ No: the total is -1 ◯ Yes: the total is 7 ◯ Yes: the total is 1 ◯ No: the total is -3

2 **One neuron can learn to say yes only when the temperature is between 18 and 26 °C.**

◯ True ◯ False

3 **Put the training of a network in order.**

☐ Repeat with the next example

☐ Nudge every weight a little downhill

☐ Measure the error against the label

☐ Guess an answer for an example

☐ Start with random weights

4 **Match each word to its meaning.**

Weight	• • • •	How the newest token weighs earlier ones
Learning rate	• • • •	Examples the model never trains on
Test set	• • • •	The size of each training step
Attention	• • • •	How much an input matters

5 **During training, a model's error on its training examples keeps falling, but its error on new examples starts to rise. What is happening?**

◯ The learning rate is too small ◯ It needs fewer examples ◯ It is learning the real pattern

◯ It is overfitting